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Exquisitor at the Video Browser Showdown 2024: Relevance Feedback Meets Conversational Search

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Abstract. An important open problem in video retrieval and exploration concerns the generation and refinement of queries for complex tasks that standard methods are unable to solve, especially when the systems are used by novices. In conversational search, the proposed approach is to ask users to interactively refine the information provided to the search process, until results are satisfactory. In user relevance feedback (URF), the proposed approach is to ask users to interactively judge query results, which in turn refines the query used to retrieve the next set of suggestions. The question that we seek to answer in the long term is how to integrate these two approaches: can the query refinements of conversational search directly impact the URF model, and can URF judgments directly impact the conversational search? We extend the existing Exquisitor URF system with conversational search. In this version of Exquisitor, the two modes of interaction are separate, but the user interface has been completely redesigned to (a) prepare for future integration with conversational search and (b) better support novice users.

Keywords: Interactive learning · Video browsing · Multimedia retrieval · Multimodal representation learning · Multimodal conversational search

1 Introduction

Video Browser Showdown (VBS) is an annual video retrieval challenge where researchers showcase their novel multimedia retrieval tools to solve complex interactive tasks. A plethora of retrieval paradigms have been seen at VBS, such as textual search [18, 26], similarity search with content-based retrieval [2, 25], sketch/object-based search [1, 25], and relevance feedback [9, 14, 27]. VBS 2024 covers three types of interactive tasks. The objective of Known Item Search (KIS) tasks is to locate a frame of a segment in a specific video. For Ad-hoc Video Search (AVS) tasks, the objective is to find as many video shots as possible that match a given description. The third task type is the Question and

Answer (Q&A) task, where the objective is to answer a question regarding the dataset, rather than submitting video clips. Another major aspect of VBS is the division into expert and novice sessions. In the expert session, each participating system’s users are typically system developers. In the novice session, the users are volunteers from the audience who are completely unfamiliar with the retrieval system. For researchers working on multimedia information retrieval, VBS is an ideal venue for real-world testing with different types of users [16, 17, 19]. Given the nature of videos, the tasks in VBS often include temporal information, and systems that introduce techniques to support temporal querying tend to perform better [1, 2, 14, 18, 22, 25]. However, in the last two editions of VBS, systems that use search with CLIP [23] features have dominated the competition, regardless of their temporal features [2, 22, 26]. There are still some tasks, however, that CLIP models struggle to solve, notably where the task does not contain much distinct information or is very domain-specific. We believe that an interactive process, where users can incrementally refine the information given to the retrieval system, will be beneficial in solving such tasks.

Exquisitor is an interactive learning system for large-scale multimedia exploration [9, 10, 13]. At its core, Exquisitor relies on user relevance feedback to build one or more classifiers using positive and negative labeled video shots determined by the user for their current information need. To support the relevance feedback process with search-oriented tasks, the user has the option to apply metadata filters that reduce the search space. Additionally, the user can search for shots using keywords (concepts, actions) or a textual query, to find better examples for the classifier. Exquisitor supports temporal querying by merging the output of two classifiers using relational operators [10] and in the most recent participation introduced a basic CLIP search implementation [13]. Based on recent VBS participation, we have observed that (a) the interface was too complex for novice users, and (b) the workflow process between search and relevance feedback was unintuitive, at times leaving users stranded in the search interface [11]. We have redesigned the Exquisitor user interface to address both aspects.

Conversational search is a method where the user starts a dialogue with a conversational agent to get answers for an information need. With advancements in image encoders and Large Language Models (LLM), it is now possible to have dialogues about multimedia content through joint visual-language models (VLM) [15]. For interactive retrieval, conversational search is a powerful tool to refine queries, by having the user provide clarifications or additional details to the replies of the agent. Unlike CLIP-based search or keyword search, where the user continuously reformulates queries, but similar to URF, conversational search works together with the user to solve the task.

In this paper, we extend the Exquisitor URF system with conversational search. Figure 1 outlines the intended workflow of the Exquisitor system, where the left side of the figure represents the flow of the conversational search and the right side represents the URF approach. The conversational flow utilizes textual input and image descriptions together with a VLM to reformulate the original query. The URF flow aims to build a classifier, using the keyframes’ semantic

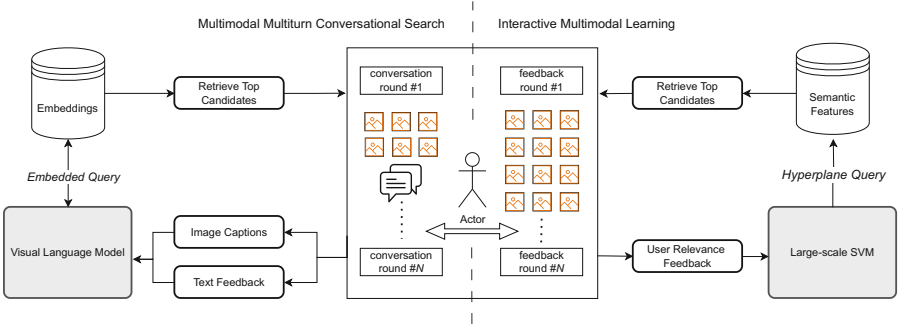


Fig. 1. Exquisitor workflow for interactive task refinement.

visual features, to acquire the top relevant keyframes from the high-dimensional index. The intention is to investigate how these two retrieval paradigms can be integrated: whether their different refinement methods can directly impact each other and what are the benefits and drawbacks of such an approach?

In the remainder of this paper, we first reevaluate Exquisitor’s current features based on observed usage and present an entirely new user interface that aims to better support conversational search and novice usage, before outlining how conversational search is integrated into the current version of Exquisitor.

2 User Relevance Feedback

2.1 Exquisitor

Exquisitor is the state-of-the-art large-scale interactive learning approach with subsecond performance on collections with over 100 million images [12, 24]. The core component of Exquisitor is an interactive learning server, which typically runs locally on a machine but can also be served for remote access. Exquisitor uses semantic features to represent multimedia items. These features are typically extracted using a deep neural network, where the resulting high-dimensional vector is compressed and stored in an approximate cluster-based high-dimensional index [6, 33]. The server has three main functions for the general user relevance feedback process: train an interactive classifier, provide suggestions, and apply filters. The interactive classifier is linear SVM; one for each modality used. Once a classifier is trained, it is used to retrieve new suggestions from the index. This retrieval process consists of traversing the index to obtain a list of clusters to consider. The items from these clusters are then ranked and the top items are returned to the user. When retrieving from clusters with active filters, query optimization policies are used to incrementally consider additional clusters until enough top items are found. If multiple modalities are used, a fused ranked list is obtained through rank aggregation [12].

The original Exquisitor UI consisted of a grid view, showing the suggestions from the underlying model, which the user could label as positive or negative.

Two containers, displaying the user’s positive and negative labeled items, were placed on the right and left sides of the suggestions. Below the suggestions was a set of buttons, which the user could use to get new sets of suggestions, view previously seen items, or reset the current model. Above the suggestions, the user could apply filters using text fields.

To perform a search, the user had to click on a search button that opened a new tab, where they could perform keyword searches, or search with CLIP, to find good examples to label as positives. The original interface featured four search types: ImageNet concepts [21], actions from Kinetics-700 [4], OCR from Tesseract and EasyOCR [8], and CLIP [23]. From participation in VBS and other similar demonstrations, it became evident that users strongly preferred CLIP search over the others. Furthermore, opening the search in a new tab disrupted the URF process. It frequently disconnected the user from the previous results, which were presented to them in the grid view. This abrupt transition often caused novice users to forget the primary purpose of the search, which was to find good examples for the classifier. Admittedly, CLIP search often returned task-relevant shots and occasionally directly solved the task at hand, but in cases where CLIP would not succeed, users would repeatedly search in vain, instead of using the returned items for relevance feedback.

2.2 Exquisitor’s User Interface

While the typical workflow was often performed well by an expert, the performance of novices varied heavily. One reason is their inexperience with interactive learning approaches, and another is the interface not being intuitive towards the workflow. In this section, we review the new Exquisitor interface, which attempts to make the interactive learning approach more intuitive, while also decluttering the screen by reducing options. In addition, the UI is made with future development in mind for the conversational search component.

Figure 2 depicts the Exquisitor interface. The top orange bar is where the user can add or remove models. A quality of life improvement for this is the addition of a dialog prompt when removing models, as in the previous interface users could accidentally hit the “x” and lose all information from a working model. The items always show the available button options for the user below it. Instead of “+” and “−” for positive and negative, we now use “Thumbs up” and “Thumbs down” buttons, which resonates better with today’s users. To submit a shot the “>” button is used, instead of the previous “S” button. At the bottom center of the grid is a hovering panel of buttons, currently only having “Update” and “Clear”. We have removed the additional buttons as they often caused confusion. While Fig. 2 shows the positive and negative containers, a major change has been to hide them initially. These containers, along with the filters from the original interface are now accessed through the buttons on the right and left sidebar. The right sidebars “Thumbs up” and “Thumbs down” buttons open containers for the positive and negative items. The history (already seen items for this model) of the current session is accessed by the “Clock” icon on the right sidebar.

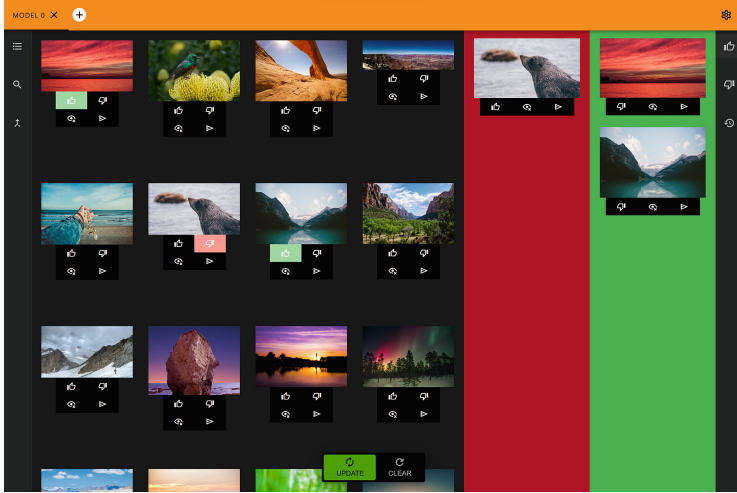


Fig. 2. The main Exquisitor display with positive and negative examples highlighted. (Color figure online)

Similar to the positive and negative containers, the filter panel can be opened by clicking the top button on the left sidebar. The main change to the filters, aside from the type of filters being more distinct, is to have the user apply the filters by pressing a button. Previously, whenever a filter was set in a field it was automatically applied, which was less transparent.

To alleviate the problems with search described above, the different search types, and the separation of search and relevance feedback, the new UI offers a panel that is accessed from the left sidebar and co-exists with the relevance feedback grid. Figure 3 shows the search panel, which is going to be the primary component for the conversational search.

3 Conversational Search

Simulating many real-world scenarios, the VBS task is not always described through a single query. Often, multiple semantic concepts, actions, or events need to be searched for before the relevant video shot is found. In addition, not all information is provided at the start when it comes to Textual Known-Item-Search tasks. Straightforward search using e.g. CLIP can form a solid base for subsequent interactive retrieval rounds. However, the user may need to rephrase or entirely reformulate the query when more information is provided or the top results are unsatisfactory. Instead of manual restarting or query reformulation, we propose a conversational search approach. The user inputs an initial query, for which a handful of the most relevant items are returned. The user is then invited to supply additional information based on the examples. They can provide textual feedback or label the items as positive or negative. We utilize the text

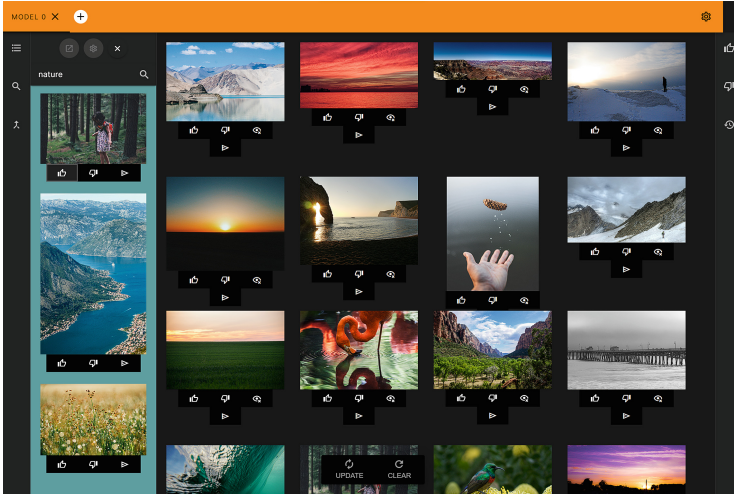


Fig. 3. The Exquisitor search panel, where conversational search will be integrated.

generation and understanding capabilities of a visual-language model (VLM) to interpret the intention of the user [3, 29]. To facilitate conversational search, we use conversational query rewriting to summarize the information contained in the multi-turn retrieval context [5, 20] and use the refined query.

Specifically, we use BLIP-2 [15] with the image encoder (ViT-B/32) to extract embeddings and LLM with Vicuna-7b to generate descriptions for each keyframe, which are stored in a database. The BLIP-2 text encoder transforms the textual query and searches the database for appropriate items. In the case of textual feedback, the VLM takes the text input and rewrites the original query. For the labeled feedback the generated descriptions of the keyframes are used as input for the VLM. We use two prompting methods for conversational search: few-shot prompting and chain-of-thought prompting (CoT). In few-shot prompting a set of examples or instructions are provided to help the VLM understand the semantic features of the target task to generate a better answer [7, 28, 31]. CoT is a step-by-step method that guides the VLM to refine the query in multi-turn conversations [30]. To make an informed choice of query refinement and retrieval approach, we deploy an automatic evaluation framework inspired by [32].

4 Conclusion

In this paper, we introduce a new version of Exquisitor, which is based on two pillars. First, the user relevance feedback of the Exquisitor system is used to refine a model to support a user’s information need through content-based feedback. Second, conversational search is used to interactively refine or build upon queries to either directly solve an information need or to provide information to enhance the relevance feedback process. The user interface of Exquisitor has been

fully redesigned based on experiences from previous participations and demonstrations to better support conversational search, and to improve the experience for novice users.

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